From Python Fundamentals to Scalable Computing: My FacultyHack@Gateways 2026 Experience

Taking part in FacultyHack@Gateways 2026 gave me a chance to rethink how I teach CSC 291: Introduction to Python Programming at Hampton University. I want my students to see Python as more than a language for small programming exercises. My curriculum project connects core Python concepts to real-world datasets, scalable computing, responsible use of Generative AI, and reproducible workflows.

The most valuable part of the experience was redesigning the course around a more authentic student project. In the revised version, students will prepare a real-world dataset, build a Python solution, and use GenAI to help with debugging, deciding for themselves whether to accept, modify, or reject each suggestion. They will then run selected workloads on Jetstream2, compare that experience with running locally or in Google Colab, and document everything as a reproducible GitHub project.

The hands-on time with HPC and research cyberinfrastructure meant a great deal to me. Working with Jetstream2 showed me how advanced computing resources can be introduced without overwhelming students. I also explored the Cornell Virtual Workshop and HPC Carpentry, which offered practical ideas for explaining HPC concepts, computational efficiency, and reproducibility in an accessible way. One lesson stood out: students don't need exposure to many platforms at once. A single, well-designed experience with a shared computing environment can be far more meaningful.

The Gateways 2026 conference also showed how researchers, educators, and gateway developers across disciplines are putting advanced computing to work. The conversations and demonstrations reinforced how important it is to lower the barriers to research computing and to bring students to these tools earlier in their academic journey.

My biggest takeaway from FacultyHack is simple: **integrate, don't just add.** HPC, AI, and science gateway technologies become far more meaningful when they connect to what students are already learning, rather than sitting apart as advanced topics.

I'm leaving with a stronger course design, more confidence using research-computing resources, and new ideas for extending these approaches into cybersecurity, data structures, AI, human-computer interaction, and undergraduate research. Most of all, I now see introductory Python as a doorway to scalable computing, responsible AI, and real-world computational problem solving.

Dr Joseph Aneke

Blog index summary (if required):
Hampton University faculty member shares how FacultyHack@Gateways 2026 inspired a redesign of CSC 291: Introduction to Python Programming. By connecting core Python concepts to real-world datasets, Jetstream2, responsible GenAI use, and reproducible workflows, the course will open the door to scalable computing from day one.